
Linear Statistical Models

B. Statistical Data Science 2nd Year Indian Statistical Institute

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Exercise Series 7

Exercise 1. In the class, we saw that in a linear model $\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\epsilon}$ with $\boldsymbol{\epsilon} \sim \mathbf{N}_n(\mathbf{0}, \sigma^2 \mathbf{I})$, the F -test statistic for testing a *testable* hypothesis $\mathcal{H}_0 : \mathbf{A}\boldsymbol{\theta} = \boldsymbol{\eta}_0$ vs. $\mathcal{H}_a : \mathbf{A}\boldsymbol{\theta} \neq \boldsymbol{\eta}_0$ is given by

$$T = \frac{(\mathbf{A}\hat{\boldsymbol{\theta}})^\top (\mathbf{A}(\mathbf{X}^\top \mathbf{X})^{-} \mathbf{A}^\top)^{-1} (\mathbf{A}\hat{\boldsymbol{\theta}})/q}{\text{SSE}/(n-k)},$$

where $\hat{\boldsymbol{\theta}} = (\mathbf{X}^\top \mathbf{X})^{-} \mathbf{X}^\top \mathbf{y}$ is a g -inverse solution to the normal equations, $k = \text{rank}(\mathbf{X})$, and q is the rank (equivalently, number of rows) of the full row-rank matrix $\mathbf{A}$. In this case, $\text{SSE} = \|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\theta}}\|^2 = \mathbf{y}^\top (\mathbf{I} - \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-} \mathbf{X}^\top) \mathbf{y}$.

We also saw that in an one-way ANOVA model

$$y_{ij} = \mu + \tau_i + \epsilon_{ij}, \quad i = 1, \dots, I, j = 1, \dots, n_i,$$

the testable hypothesis $\mathcal{H}_0 : \tau_1 = \dots = \tau_I$ can be written as $\mathcal{H}_0 : \mathbf{A}\boldsymbol{\theta} = \mathbf{0}$, where

$$\boldsymbol{\theta} = \begin{pmatrix} \mu \\ \tau_1 \\ \vdots \\ \tau_I \end{pmatrix} \quad \text{and} \quad \mathbf{A} = \begin{pmatrix} 0 & 1 & -1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & -1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & 1 & -1 \end{pmatrix}.$$

We also saw that with this formulation, it is not easy to find the inverse of the matrix $\mathbf{A}(\mathbf{X}^\top \mathbf{X})^{-} \mathbf{A}^\top$.

- (a) Verify that $\mathcal{H}_0 : \tau_1 = \dots = \tau_I$ can be alternatively formulated as $\mathcal{H}_0 : \begin{matrix} \tau_1 = \tau_2 \\ \tau_1 = \tau_3 \\ \vdots \\ \tau_1 = \tau_I \end{matrix}$. Write this in

vector-matrix notation as $\mathcal{H}_0 : \mathbf{C}\boldsymbol{\theta} = \mathbf{0}$.

- (b) Find $\mathbf{C}(\mathbf{X}^\top \mathbf{X})^{-} \mathbf{C}^\top$ for this formulation. Show that it is of the form $\mathbf{D} + \mathbf{u}\mathbf{v}^\top$, where $\mathbf{D}$ is a diagonal matrix.

Hint: Recall that $(\mathbf{X}^\top \mathbf{X})^{-} = \begin{pmatrix} 0 & 0 & 0 & \dots & 0 \\ 0 & 1/n_1 & 0 & \dots & 0 \\ 0 & 0 & 1/n_2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1/n_I \end{pmatrix}$ is a g -inverse of $\mathbf{X}^\top \mathbf{X}$ in this case.

- (c) Use the previous part to find $(\mathbf{C}(\mathbf{X}^\top \mathbf{X})^{-} \mathbf{C}^\top)^{-1}$.

Hint: Series 2, Exercise 4(b).

- (d) Use the previous part to show that $(\mathbf{C}\hat{\boldsymbol{\theta}})^\top (\mathbf{C}(\mathbf{X}^\top \mathbf{X})^{-} \mathbf{C}^\top)^{-1} (\mathbf{C}\hat{\boldsymbol{\theta}}) = \sum_{i=1}^I n_i (\bar{y}_{i.} - \bar{y}_{..})^2$.

- (e) Verify that $X\hat{\boldsymbol{\theta}} = (\underbrace{\bar{y}_{1.}, \dots, \bar{y}_{1.}}_{n_1}, \dots, \underbrace{\bar{y}_{I.}, \dots, \bar{y}_{I.}}_{n_I})^\top$. Hence, show that $\text{SSE} = \sum_{i=1}^I \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{i.})^2$.
- (f) Use (d) and (e) to conclude that the F -test statistic obtained using this direct approach coincides with the statistic obtained using the full-reduced model approach in the class.

Note: We already (theoretically) knew that the test statistics obtained using the different approaches coincide. This is a verification of that result.

Exercise 2. We have seen that the same set of hypotheses can be formulated in different ways. For instance, $\mathcal{H}_0 : \tau_1 = \dots = \tau_I$ in a one-way ANOVA model can be formulated equivalently as $\mathcal{H}'_0 : A\boldsymbol{\theta} = \mathbf{0}$ as well as $\mathcal{H}''_0 : B\boldsymbol{\theta} = \mathbf{0}$, where

$$A = \begin{pmatrix} 0 & 1 & -1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & -1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & 1 & -1 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 0 & 1 & -1 & 0 & \dots & 0 & 0 \\ 0 & 1 & 0 & -1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 1 & 0 & 0 & \dots & 0 & -1 \end{pmatrix}.$$

In our following discussion, we assume that the matrices A and B are of full row-rank. This ensures that there is no redundancy in our hypothesis formulation.

- (a) Let $\mathcal{H}_0 : A\boldsymbol{\theta} = \mathbf{0}$ and $\mathcal{H}'_0 : B\boldsymbol{\theta} = \mathbf{0}$ be two hypotheses. Show that $\mathcal{H}_0$ and $\mathcal{H}'_0$ are equivalent if and only if $\mathcal{R}(A) = \mathcal{R}(B)$. This is ensured if and only if $A = VB$ for some non-singular (square) matrix V .
- Note:** $\mathcal{H}_0$ and $\mathcal{H}'_0$ are equivalent means that $\mathcal{H}_0$ implies $\mathcal{H}'_0$ and vice versa. Thus, there is no difference in testing either of the hypotheses.
- (b) Use part (a) to show that the F -test statistics for testing two equivalent hypotheses $\mathcal{H}_0 : A\boldsymbol{\theta} = \mathbf{0}$ and $\mathcal{H}'_0 : B\boldsymbol{\theta} = \mathbf{0}$ are the same. Thus, the conclusion of the F -test does not depend on the hypothesis formulation.

Exercise 3 (Practical 2, Graded). The `cigarette` data set contains the diastolic blood pressures* of several individuals along with their ages and smoking preferences. Our goal is to model the blood pressure using covariates `age` and `smoker`.

- (a) Plot `pressure` against `age`. Do you think a linear function in `age` is suitable to model `pressure` here? Now, make the same plot, but separately for “smokers” and “non-smokers”. What do you think about the suitability of a linear model now?
- (b) Fit a linear model for `pressure` on `age` only. What do the parameters of this model signify? Plot the fitted model on the scatterplot of the data. Is it a good fit?
- (c) Now, add the variable `smoker` to your model. What do the parameters of this model signify? Try to visualize the fitted model by plotting the observed and fitted values of `pressure` against `age` on the same graph. Compare this with the previous model in terms of its fit.
- (d) Finally, we add an interaction term to the model. You can use `age:smoker` in `lm` to indicate an interaction term (product term) involving `age` and `smoker`. What do the parameters of this model signify? Visualize the model similarly to the previous part. What do you think about this model in terms of its fit?
- (e) As we have discussed in the class, the variable `smoker` in this data set is a **categorical** covariate with two categories – `yes` and `no`. As such, there is no reason to give preference to either of

this categories, and our model stays the same (except for interpretations of the parameters) if we interchange the categories. We will verify this numerically. Interchange the tags “yes” and “no” for `smoker` and refit the models in parts (d) and (e). Do you see any connection between the estimated parameters of these models and the models in (d) and (e)? Also, do you see how `lm` handles categorical covariates with two categories?

- (f) Check that the estimates of parameters of the refitted models could be obtained directly from the estimated parameters of the former models, without refitting. Obtain the corresponding formulæ.

* Don't take the numbers seriously, this is a simulated data set.

Exercise 4. Let A and B be two factor variables with factors $A_1, \dots, A_I$ and $B_1, \dots, B_J$, respectively. Consider a two-way ANOVA model with one observation per cell $y_{ij} = \mu_{ij} + \epsilon_{ij}, i = 1, \dots, I, j = 1, \dots, J$, where μ_{ij} is the effect due to the simultaneous application of A_i and B_j . Suppose we consider an additive model: $\mu_{ij} = \mu + \tau_i + \beta_j + \gamma_{ij}$, as discussed in the class.

- (a) Find the degrees of freedom of SSE in this case.
- (b) Comment on the feasibility of the interaction term in the model.

Exercise 5 (Practical 3). The time series `AirPassengers` provides the monthly totals of international airline passengers from 1949 to 1960.

- (a) Fit a linear model linking the number of passengers to the time of the year. Add a monthly effect to your model by considering `month` as a factor variable. How does it affect the fit? Use the model to predict the number of passengers in December 1962.
- (b) Make diagnostic plots. You may use `plot` on a fitted linear model in R. Do you notice anything?
- (c) The plot of `passengers` on `year` suggests that the growth in aerial traffic is exponential during the time period under study. Try to fit a linear model with the log of the number of passengers as response. Is this a better fit than the previous models?
- (d) Plot the lagged residuals. Do you think all the assumption in linear models are justified?